

Problem: (15 points)

Johnny is deciding whether to pursue a **three-year college diploma**. He evaluates this decision purely as an **investment in human capital**.

- Length of program: **3 years**
- Annual direct costs:
 - Tuition: \$1,200
 - Books and materials: \$300
- If Johnny does **not** attend college: – He can earn \$8,000 per year working immediately
- If Johnny **completes** the diploma: – He will earn \$14,000 per year
- Time horizon: – Johnny will work for **12 years after graduation**
- Market interest rate: $r = 6\%$

Questions

1. Calculate Johnny's **total annual cost of attending college**.
2. Compute the **opportunity cost per year** of attending college.
3. Calculate the **total cost of the investment** over the 3-year period (ignore discounting).
4. Calculate the **annual earnings gain** from completing the diploma.
5. Compute the **present value of benefits** (additional earnings after graduation).
6. Compute the **present value of costs** (including both direct and opportunity costs).
7. Should Johnny invest in the diploma based on the **net present value** rule?
8. Suppose the market interest rate rises to 8%. Explain how this affects Johnny's decision.
9. Determine approximately the **internal rate of return (IRR)** of the investment.
10. Explain how Johnny's decision would change if:
 - (a) His expected post-education earnings fall
 - (b) Tuition fees increase
 - (c) His working life becomes shorter